CLASS 10 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The point which divides $(6, 0)$ and $(0, 10)$ in the ratio 1:1 has coordinates summing to: 1

(a) 16 **(b)** 8 **(c)** 4 **(d)** 30

2. If $\sin\theta = \frac{4}{5}$, then $\sec\theta + \tan\theta =$ 1

(a) $\frac{1}{3}$ **(b)** $\frac{5}{3}$ **(c)** 3 **(d)** $\frac{9}{4}$

3. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 8 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) $\frac{8\sqrt{3}}{3}$ m **(b)** $8\sqrt{3}$ m **(c)** 4 m **(d)** $4\sqrt{3}$ m

4. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 30 cm. The length of each tangent is 16 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 92 cm **(b)** 136 cm **(c)** 46 cm **(d)** 32 cm

5. The largest possible circle is cut from a square sheet of side 56 cm. The area of the sheet left over is: 1

(a) 3136 cm² **(b)** 672 cm² **(c)** 1344 cm² **(d)** 2464 cm²

6. A hemispherical bowl has outer radius 6 cm and inner radius 3 cm. The volume of material used to make the bowl is: 1

(a) 1188 cm³ **(b)** 198 cm³ **(c)** 396 cm³ **(d)** 792 cm³

7. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 10, 6, 15, 12, 5, 2 respectively, The median of the data is: 1

(a) 26 **(b)** 25.4 **(c)** 20 **(d)** 30

8. A coin is tossed three times. The probability of getting at least one tail is: 1

(a) $\frac{7}{8}$ **(b)** $\frac{3}{4}$ **(c)** 1 **(d)** $\frac{1}{8}$

9. The largest number that divides 70 and 125 leaving remainders 5 and 8 respectively is: 1

(a) 65 **(b)** 26 **(c)** 13 **(d)** 1

10. If $x = 2$ is a zero of $x^2 - 7x + k$, then k is: 1

(a) 14 **(b)** 10 **(c)** 18 **(d)** 5

11. The pair $2x + ky = 6$, $4x + 4y = 12$ has infinitely many solutions when $k =$ 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 6

12. The roots of $x^2 - 11x + 18 = 0$ differ by: 1

(a) 18 **(b)** 7 **(c)** 11 **(d)** 8

13. The sum of all two-digit multiples of 3 is: 1

(a) 1665 **(b)** 1566 **(c)** 1683 **(d)** 1653

14. The altitude to the hypotenuse of a right triangle divides it into 4 cm and 9 cm. The altitude (cm) is: 1

(a) 36 **(b)** $\frac{13}{2}$ **(c)** 6 **(d)** 13

15. The centroid of the triangle with vertices $(1, 2)$, $(4, 6)$, $(7, 1)$ has x-coordinate: 1

(a) 6 **(b)** 4 **(c)** 12 **(d)** 3

16. If $\cos\theta = \frac{3}{5}$, then $\operatorname{cosec}\theta - \cot\theta =$ 1

(a) $\frac{5}{4}$ **(b)** $\frac{1}{2}$ **(c)** 2 **(d)** $\frac{1}{3}$

17. Two points on level ground lie on opposite sides of a tower of height $6\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. The distance between the two points is: 1

(a) 12 m **(b)** $24\sqrt{3}$ m **(c)** 48 m **(d)** 24 m

18. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 40 cm. The length of each tangent is 9 cm. The area of the quadrilateral $PAOB$ (in cm^2) is: 1

(a) 98 cm² **(b)** 180 cm² **(c)** 720 cm² **(d)** 360 cm²

19. A circular park of radius 20 m is surrounded by a path 7 m wide. The area of the path is: 1

(a) 880 m² **(b)** 517 m² **(c)** 1034 m² **(d)** 2068 m²

20. A hemispherical depression of diameter 21 cm, the largest possible, is scooped out of the top face of a cubical block of edge 21 cm. The volume of the remaining block is: 1

(a) 6835.5 cm³ **(b)** 4410 cm³ **(c)** 2425.5 cm³ **(d)** 9261 cm³

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Find the values of y for which the distance between $P(7,\ y)$ and $Q(1,\ 2)$ is 10 units. 2

22. If $\tan A = \frac{4}{3}$, find $\dfrac{\sin A + \cos A}{\sin A - \cos A}$. 2

23. From the top of a building 6 m high, the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 60°. Find the height of the tower. 2

24. PA and PB are tangents from an external point P to a circle, and $\angle PAB = 30^\circ$. Find the $\angle APB$. 2

25. The minute hand of a clock is 28 cm long. Find the area swept by the minute hand in 15 minutes. 2

26. A solid metallic cone of base radius 6 cm and height 3 cm is melted and recast into a solid sphere. Find the radius of the sphere. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. The angle of elevation of the top of a vertical tower from a point on level ground is 45°. After walking 4 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. **3**

28. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (9x + 4)$ cm and $QB = (5x + 20)$ cm. Find the length of QA. **3**

29. A circular field has radius 14 m. Fencing costs ₹ 12 per metre. Find the cost (in ₹) of fencing the field once. **3**

30. A capsule is a cylinder of radius 10.5 cm and length 12 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. **3**

31. The marks obtained by students in a test are given below. **3**

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	10	6	15	12	5	2

Find the mean.

OR

A bag contains 7 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. Find the value of x. **3**

32. Three bells toll at intervals of 12, 15 and 18 minutes. If they toll together at 8:00 a.m., after how many minutes will they next toll together? **3**

OR

Find a quadratic polynomial whose zeroes are the reciprocals of the zeroes of $x^2 - 5x + 6$. **3**

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Tangents PA and PB are drawn from a point P at a distance of 52 cm from the centre O of a circle of radius 20 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). **5**

34. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 45° is marked in both. Find the area of the region between the two arcs within the sector. **5**

35. A conical vessel of radius 6 cm and height 20 cm is full of water. The water is poured into a cylindrical vessel of base radius 4 cm. Find the height of the water in the cylindrical vessel. **5**

36. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 and frequencies 6, 9, 7, p, 16, 7 respectively is 32.4. Find the value of p. **5**

OR

Two dice are thrown together. Find the probabilities of these events (in this order): **5**
(i) the product of the numbers is 12,
(ii) the sum is a prime number,
(iii) the sum is a multiple of 3.

37. A rectangular courtyard is 24 m by 16 m. Find the side of the largest square tile that fits exactly, and the number of tiles. **5**

OR

Find k if the sum of the zeroes of $kx^2 - 12x + 8$ is 3, and hence find the product of its zeroes. 5

38. A fraction becomes $\frac{1}{2}$ when 1 is added to both the numerator and the denominator, and becomes $\frac{5}{9}$ when 2 is added to both. Find the fraction. 5

OR

The altitude of a right triangle is 1 cm less than its base and the hypotenuse is 29 cm. Find the base and altitude. 5

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND